TRIGONOMETRY

(त्रिकोणमिति)

SHEET – 05

CLASS ROOM SHEETS / FOR ALL EXAMS

UPDATED

TRIGONOMETRY (त्रिकोणमिति)

CLASSROOM SHEET-05

Multiple and Sub-multiple Trigonometric function of 2A

2A के गुणक तथा उपगुणक त्रिकोणमितीय फलन

- $\sin 2A = 2\sin A.\cos A = \dfrac{2\tan A}{1+\tan^2 A}$
- $\sin A = 2\sin\dfrac{A}{2}\cos\dfrac{A}{2} = \dfrac{2\tan\frac{A}{2}}{1+\tan^2\frac{A}{2}}$
- $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1$
  $= 1 - 2\sin^2 A = \dfrac{1-\tan^2 A}{1+\tan^2 A}$
- $\cos A = \cos^2\dfrac{A}{2} - \sin^2\dfrac{A}{2} = 2\cos^2\dfrac{A}{2} - 1$
  $= 1 - 2\sin^2\dfrac{A}{2} = \dfrac{1-\tan^2\frac{A}{2}}{1+\tan^2\frac{A}{2}}$
- $1 - \cos A = 2\sin^2\dfrac{A}{2}$
- $1 + \cos A = 2\cos^2\dfrac{A}{2}$
- $\tan 2A = \dfrac{2\tan A}{1-\tan^2 A}$
- $\tan A = \dfrac{2\tan\frac{A}{2}}{1-\tan^2\frac{A}{2}}$
- $\cot 2A = \dfrac{\cot^2 A - 1}{2\cot A}$

Trigonometric function of 3A

3A के त्रिकोणमितीय फलन

- $\sin 3A = 3\sin A - 4\sin^3 A$
- $\cos 3A = 4\cos^3 A - 3\cos A$
- $\tan 3A = \left(\dfrac{3\tan A - \tan^3 A}{1-3\tan^2 A}\right)$
- $\cot 3A = \left(\dfrac{\cot^3 A - 3\cot A}{3\cot^2 A - 1}\right)$

1. If $2(\cos^2\theta - \sin^2\theta) = 1$ (θ is a positive acute angle), then $\cot\theta$ is equal to :

(a) $-\sqrt{3}$ (b) $\dfrac{1}{\sqrt{3}}$

(c) 1 (d) $\sqrt{3}$

2. The value of

$\left(\dfrac{2\cos^3\theta - \cos\theta}{\sin\theta - 2\sin^3\theta}\right)^2 + 1, \theta \neq 45°$ is:

SSC CPO 28/06/2024 (Shift-03)

(a) $\sin^2\theta$ (b) $\sec^2\theta$

(c) $\text{cosec}^2\theta$ (d) $\cot^2\theta$

3. Simplify the following expression

$\dfrac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta}$

SSC CGL 21/07/2023 (Shift-03)

(a) $\tan\theta$ (b) $\sin\theta$

(c) $\sec\theta$ (d) $\cos\theta$

4. Evaluate the following:

$\sqrt{2+\sqrt{2+\sqrt{2+2\cos 8\theta}}}$

SSC CGL 27/07/2023 (Shift-3)

(a) $2\cos\theta$ (b) $2\cos 2\theta$

(c) $\sin 2\theta$ (d) $\cos 2\theta$

5. If $4\tan\theta - 3 = 0$, then the value of $\dfrac{1-\cos 2\theta}{1+\cos 2\theta}$ is?

SSC CGL 23/09/2024 (Shift-01)

(a) 1 (b) $\dfrac{7}{15}$

(c) $\dfrac{4}{3}$ (d) $\dfrac{9}{16}$

6. $\dfrac{2\tan A}{1+\tan^2 A} = ?$

SSC CGL 20/07/2023 (Shift-02)

(a) cos2A (b) sinA

(c) cosA (d) sin2A

7. If $\cos^4\theta - \sin^4\theta = \dfrac{4}{5}$, then find the value of $\sin 4\theta$.

SSC CHSL 08/07/2024 (Shift-04)

(a) $\dfrac{24}{25}$ (b) $\dfrac{21}{25}$

(c) $\dfrac{16}{25}$ (d) $\dfrac{18}{25}$

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8. The value of $\frac{\sin 4\theta}{(1-\cos 4\theta)}$ is:

SSC CHSL 01/06/2022 (Shift- 03)

(a) cot θ (b) cot 2θ

(c) tan θ (d) tan 2θ

9. What is the value of $\cos^2 15°$?

SSC CGL 06/12/2022 (Shift- 04)

(a) $(2 + \sqrt{3})$ (b) $\frac{(2+\sqrt{3})}{4}$

(c) $\frac{(2+\sqrt{3})}{2}$ (d) $\frac{(1+\sqrt{3})}{2}$

10. What is the value of $\text{cosec}15° \sec 15°$

SSC CGL 13/12/2022 (Shift- 02)

(a) 0.5 (b) 4

(c) 2 (d) 1

11. What is the value of the expression $100(\sin 15° \cos 15°)$?

SSC CHSL 03/06/2022 (Shift- 03)

(a) 50 (b) 75

(c) 100 (d) 25

12. Find the value of $\frac{1-\tan^2 22\frac{1}{2}^\circ}{1+\tan^2 22\frac{1}{2}^\circ}$ is :

(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{1}{\sqrt{2}}$

(c) $\frac{1}{2}$ (d) $\sqrt{3}$

13. Find the value of $3\sin 15° - 4\sin^3 15°$.

SSC CGL 25/09/2024 (Shift-03)

(a) 2 (b) $\frac{1}{2}$

(c) $\frac{1}{\sqrt{2}}$ (d) $\sqrt{2}$

14. Find the value of $\sqrt{\frac{1-\sin 3\theta}{1+\sin 3\theta}}$

SSC CGL TIER I 21/07/2023 (Shift-02)

(a) $\sec 3\theta - \tan 3\theta$ (b) $(\sec 3\theta - \tan 3\theta)^2$

(c) $(\sec 3\theta - \tan 3\theta)^3$ (d) $\sec 3\theta + \tan 3\theta$

15. What is the value of $(4\sin^3 x - 3\sin x + \sin 3x)$?

SSC CHSL 31/05/2022 (Shift- 02)

(a) 1 (b) 0

(c) 4 (d) 3

16. If $\sin\theta = \frac{\sqrt{3}}{2}$, and θ is an acute angle, find the value of $\cos 3\theta$.

SSC CHSL 10/06/2022 (Shift- 01)

(a) $\frac{1}{2}$ (b) 1

(c) – 1 (d) 0

17. If $\sin\theta = \frac{1}{2}$, then the value of $(3\cos\theta - 4\cos^3\theta)$ is:

SSC CGL 06/12/2022 (Shift- 03)

(a) 0 (b) 1

(c) 2 (d) –1

18. What will be the value of $\sin 10° - \frac{4}{3}\sin^3 10°$?

SSC CGL 06/12/2022 (Shift-01)

(a) $\frac{1}{3\sqrt{3}}$ (b) $\frac{1}{6}$

(c) $\frac{1}{2\sqrt{3}}$ (d) $\frac{\sqrt{3}}{6}$

Identities based on Sum & Difference of angle of T-Ratio
त्रिकोणमिति अनुपात के कोणों का योग तथा अंतर पर आधारित सर्वसमिकाएँ

- $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- $\sin(A - B) = \sin A \cos B - \cos A \sin B$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B$
- $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
- $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$
- $\cot(A + B) = \frac{\cot A \cot B - 1}{\cot B + \cot A}$
- $\cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$

19. Which of the following options gives an expression equivalent to sin(A + B)?

SSC CGL 24/07/2023 (Shift-2)

(a) cosAcosB – sinAsinB

(b) sinAcosB + cosAsinB

(c) cosAcoSB + sinAsinB

(d) sinAcosB-cosAsinB

20. If $\sin A = \frac{4}{5}$ and $\sin B = \frac{15}{17}$, what is the value of sin(A – B)?

SSC CGL 02/12/2022 (Shift- 04)

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(a) $-\frac{32}{45}$ (b) $-\frac{13}{85}$

(c) $\frac{13}{85}$ (d) $\frac{32}{45}$

21. What is the value of cos45° sin15°?

SSC CGL 07/12/2022 (Shift- 02)

(a) $\frac{\left(\sqrt{3}-1\right)}{2}$ (b) $\frac{\left(\sqrt{3}-1\right)}{4}$

(c) $(\sqrt{3}+1)$ (d) $2\sqrt{3}-1$

22. What is the value of sin75° + sin15°?

SSC CGL 09/12/2022 (Shift- 04)

(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{\sqrt{3}}{2}$

(c) $\sqrt{\frac{3}{2}}$ (d) $\frac{3}{\sqrt{2}}$

23. Evaluate the following

$\sin25^\circ \sin65^\circ - \cos25^\circ \cos65^\circ$.

SSC CGL 14/07/2023 (Shift-2)

(a) 40 (b) 4

(c) 0 (d) 1

24. Using cos (A + B) = cosA cosB – sin A sin B, find the value of cos75°

SSC CGL 17/07/2023 (Shift-03)

(a) $\frac{\sqrt{5}-1}{4}$ (b) $\frac{\sqrt{5}+1}{4}$

(c) $\frac{\sqrt{6}-\sqrt{2}}{4}$ (d) $\frac{\sqrt{6}+\sqrt{2}}{4}$

25. What is the value of cos $(x + y)$–sin $(x - y)$ + tan $(2z)$?

SSC CHSL 02/07/2024 (Shift-01)

(a) cos x. cos y – sin x. sin y – sin x. cos y – cos x . sin y + $\frac{2\tan z}{1-\tan^2 z}$

(b) cos x. cos y – sin x. sin y – sin x. cos y + cos x. sin y + $\frac{2\tan z}{1-\tan^2 z}$

(c) cos x. cos y – sin x. sin y – sin x. cos y + cos x. sin y + $\frac{\tan z}{1-\tan^z z}$

(d) sin x. sin y + $\frac{2\tan z}{1-\tan^z z}$

26. Using tan (A – B) = $\frac{\tan A - \tan B}{1+\tan A \tan B}$, find the value of tan 15°.

SSC CGL 18/07/2023 (Shift-04)

(a) $\sqrt{3}+1$ (b) $\sqrt{3}-1$

(c) $2-\sqrt{3}$ (d) $2+\sqrt{3}$

27. What will be the value of $\frac{\tan45^\circ - \tan15^\circ}{1+\tan45^\circ \tan15^\circ}$?

SSC CGL 07/12/2022 (Shift-03)

(a) $\sqrt{3}$ (b) $\frac{1}{\sqrt{2}}$

(c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{2}$

28. Find the value of tan 72° – tan 27° – tan 72° tan 27°.

SSC CGL 25/09/2024 (Shift-02)

(a) –1

(b) 1

(c) –2

(d) 0

29. Simplify: $\frac{\cos13^\circ + \sin13^\circ}{\cos13^\circ - \sin13^\circ}$

SSC CPO 28/06/2024 (Shift-01)

(a) cot 58° (b) tan 32°

(c) tan 58° (d) cos 26°

30. If tan(α + β) = a, tan(α – β) = b, then the value of tan2α is:

SSC CGL 13/12/2022 (Shift-04)

(a) $\frac{a+b}{1-ab}$ (b) $\frac{a+b}{1+ab}$

(c) $\frac{a-b}{1+ab}$ (d) $\frac{a-b}{1-ab}$

31. Using cot (A – B) = $\frac{\cot A \cot B + 1}{\cot B - \cot A}$, find the value of cot15°.

SSC CGL 19/07/2023 (Shift-02)

(a) $2+\sqrt{3}$ (b) $2-\sqrt{3}$

(c) $\sqrt{3}-1$ (d) $\sqrt{3}+1$

32. If sin(A + B) = cos(A + B), what is the value of tanA?

SSC CGL 02/12/2022 (Shift- 01)

(a) $\frac{1-\tan B}{1+\tan B}$ (b) $\frac{1+\tan B}{1-\tan B}$

(c) $\frac{1+\sec B}{1-\sec B}$ (d) $\frac{1-\operatorname{cosec} B}{1+\operatorname{cosec} B}$

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Transformation of sum or difference into product
योग अथवा अन्तर का गुणन में परिवर्तन

- $\sin C + \sin D = 2\sin\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$
- $\sin C - \sin D = 2\cos\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$
- $\cos C + \cos D = 2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$
- $\cos C - \cos D = 2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{D-C}{2}\right)$

33. If sinC + sinD = x, then the value of x is :

(a) $2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

(b) $2\sin\left(\frac{C-D}{2}\right)\cos\left(\frac{C+D}{2}\right)$

(c) $2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

(d) $2\sin\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

34. What is $\sin\alpha - \sin\beta$?

SSC CGL 01/12/2022 (Shift- 02)

(a) $2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$

(b) $2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$

(c) $2\cos\frac{\alpha-\beta}{2}\sin\frac{\alpha+\beta}{2}$

(d) $2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$

35. If cosC – cosD = y, then the value of y is :

(a) $2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

(b) $2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

(c) $-2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

(d) $-2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

36. Find the value of:
(sin 75° + sin 15°).

SSC Phase-XII 21/06/2024 (Shift-03)

(a) $\frac{\sqrt{6}}{2}$ (b) $\sqrt{6}$

(c) $\frac{\sqrt{3}}{2}$ (d) $\sqrt{3}$

37. Find $\frac{\cos 7x + \cos 5x}{\sin 7x - \sin 5x}$ = ?

(a) cot2x (b) cot3x
(c) cot$x$ (d) cot6x

38. Find $\frac{\sin 5x - 2\sin 3x + \sin x}{\cos 5x - \cos x}$ = ?

(a) tan4x (b) tan3x
(c) tan2x (d) tan$x$

39. What is the value of

$\frac{[(\sin x + \sin y)(\sin x - \sin y)]}{[(\cos x + \cos y)(\cos x - \cos y)]}$ is :

(a) 0 (b) 1
(c) – 1 (d) 2

40. What is the value of sin75° + sin15° ?

(a) $\sqrt{3}$ (b) $\frac{2}{\sqrt{3}}$

(c) $\sqrt{\frac{3}{2}}$ (d) $\frac{3}{\sqrt{2}}$

41. Simplify the following:
sin 2x + 2sin4x + sin 6x

SSC CGL 18/07/2023 (Shift-02)

(a) 4 $\cos^2 x \sin 4x$ (b) 4 $\cos^2 x \sin x$
(c) 2 $\cos^2 x \sin 4x$ (d) 4 $\sin^2 x \sin 4x$

Transformation of a product into sum or difference
गुणन का योग या अन्तर में परिवर्तन

(i) $\sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$
(ii) $\cos(A+B)\cos(A-B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 B$
(iii) $2\sin A\cos B = \sin(A+B) + \sin(A-B)$
(iv) $2\cos A\sin B = \sin(A+B) - \sin(A-B)$
(v) $2\cos A\cos B = \cos(A+B) + \cos(A-B)$
(vi) $2\sin A\sin B = \cos(A-B) - \cos(A+B)$

42. Using 2cosA cosB = cos(A + B) + cos(A – B), find the value of cos75°cos15°.

SSC CHSL 01/06/2022 (Shift- 02)

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$

(c) $\frac{\sqrt{3}}{2}$ (d) $\frac{\sqrt{3}}{4}$

43. What will be value of 2cos45°. sin15°

(a) $\frac{\sqrt{3}+1}{2}$ (b) $\frac{\sqrt{3}-2}{2}$

(c) $\frac{\sqrt{3}-1}{2}$ (d) $\frac{2}{\sqrt{3}+1}$

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44. Simplify: $\frac{\sin 8\theta\cos\theta - \sin 6\theta\cos 3\theta}{\cos 2\theta\cos\theta - \sin 3\theta\sin 4\theta}$

SSC CPO 09/11/2022 (Shift-02)

(a) $\cot\theta$ (b) $\cot 2\theta$
(c) $\tan\theta$ (d) $\tan 2\theta$

45. Evaluate the following:
$\cos(36° + A).\cos(36° - A) + \cos(54° + A).\cos(54° - A)$

SSC CGL 14/07/2023 (Shift-2)

(a) $\sin 2A$ (b) $\cos A$
(c) $\sin A$ (d) $\cos 2A$

46. If A + B = C, then tanA tanB tanC = ?

SSC CGL 08/12/2022 (Shift- 01)

(a) tanC + tanA – tanB
(b) tanC + tanA + tanB
(c) tanA – tanB – tanC
(d) tanC – tanA – tanB

47. What is the value of the expression cos2A cos2B + $\sin^2(A - B) - \sin^2(A + B)$?

SSC CGL 12/12/2022 (Shift- 01)

(a) sin (2A – 2B) (b) sin (2A + 2B)
(c) cos (2A + 2B) (d) cos (2A – 2B)

48. If cot 75° = 2 – $\sqrt{3}$. Find the value of cot15°.

SSC CHSL 24/05/2022 (Shift- 03)

(a) $2 - \sqrt{3}$ (b) $2 + \sqrt{3}$
(c) $\sqrt{3} + 1$ (d) $\sqrt{3} - 1$

➢ If A + B = 45° = 225°,

(i) (1 + tanA) (1 + tanB) = 2

(ii) (1 – cotA) (1 – cotB) = 2

49. The value of (1 + tan 10°) (1 + tan 35°) is:

SSC CHSL 25/05/2022 (Shift- 03)

(a) $\frac{1}{2}$
(b) $\frac{3}{4}$
(c) 1
(d) 2

50. $\left[1+\tan\left(22\frac{1}{2}^\circ + x - y\right)\right]\left[1+\tan\left(22\frac{1}{2}^\circ + y - x\right)\right]$

(a) 0
(b) 1
(c) 2
(d) 4

51. Find the value of (1 + tan11°) (1 + tan 21°) (1 + tan 24°) (1 + tan 34°) (1 + tan 45°)

(a) 0 (b) 1
(c) 2 (d) 8

Some Important Results
कुछ महत्वपूर्ण परिणाम

(i) $\sin A.\sin 2A.\sin 4A = \frac{1}{4}\sin 3A$

(ii) $\cos A.\cos 2A.\cos 4A = \frac{1}{4}\cos 3A$

(iii) $\tan A.\tan 2A.\tan 4A = \tan 3A$

(iv) $\sin(60° - A).\sin A.\sin(60° + A) = \frac{1}{4}\sin 3A$

(v) $\cos(60° - A).\cos A.\cos(60° + A) = \frac{1}{4}\cos 3A$

(vi) $\tan(60° - A).\tan A.\tan(60° + A) = \tan 3A$

(vii) $\sin^2(60° - A) + \sin^2 A + \sin^2(60° + A) = \frac{3}{2}$

(viii) $\cos^2(60° - A) + \cos^2 A + \cos^2(60° + A) = \frac{3}{2}$

52. Find sin20°.sin40°.sin60°.sin80°.

(a) $\frac{1}{16}$ (b) $\frac{5}{16}$
(c) $\frac{3}{16}$ (d) $\frac{1}{8}$

53. The value of (tan20°.tan40°.tan80°) is :

(a) 1 (b) 0
(c) $\sqrt{3}$ (d) 3

54. Find 1 – sin10°.sin50°.sin70°.

(a) $\frac{5}{8}$ (b) $\frac{7}{8}$
(c) $\frac{3}{8}$ (d) $\frac{1}{8}$

55. sin6° sin42° sin66° sin78° = ?

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{1}{8}$ (d) $\frac{1}{16}$

56. Find sin12°.sin48°.sin54°.

(a) $\frac{5}{8}$ (b) $\frac{7}{8}$
(c) $\frac{3}{8}$ (d) $\frac{1}{8}$

57. The value of (cos15°.cos45°.cos75°) is :

(a) $\frac{1}{3\sqrt{2}}$ (b) $\frac{1}{4\sqrt{2}}$
(c) $\frac{1}{8}$ (d) $\frac{\sqrt{3}}{8}$

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58. $\cos10^\circ.\cos30^\circ.\cos50^\circ.\cos70^\circ = ?$

(a) $\frac{1}{16}$ (b) $\frac{3}{16}$

(c) $\frac{5}{16}$ (d) $\frac{1}{8}$

59. $\cos42^\circ.\cos78^\circ.\cos36^\circ = ?$

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$

(c) $\frac{1}{8}$ (d) $\frac{1}{16}$

60. The value of $\tan6^\circ.\tan42^\circ.\tan66^\circ.\tan78^\circ$ is :

(a) 1

(b) $\frac{1}{2}$

(c) $\frac{1}{4}$

(d) $\frac{1}{8}$

ANSWER KEY

1.	(d)	2.	(a)	3.	(a)	4.	(a)	5.	(d)	6.	(d)	7.	(a)	8.	(b)	9.	(b)	10.	(b)
11.	(d)	12.	(b)	13.	(c)	14.	(a)	15.	(b)	16.	(c)	17.	(a)	18.	(b)	19.	(b)	20.	(b)
21.	(b)	22.	(c)	23.	(c)	24.	(c)	25.	(b)	26.	(c)	27.	(c)	28.	(b)	29.	(c)	30.	(a)
31.	(a)	32.	(a)	33.	(d)	34.	(a)	35.	(c)	36.	(a)	37.	(c)	38.	(d)	39.	(c)	40.	(c)
41.	(a)	42.	(b)	43.	(c)	44.	(d)	45.	(d)	46.	(d)	47.	(c)	48.	(b)	49.	(d)	50.	(c)
51.	(d)	52.	(c)	53.	(c)	54.	(b)	55.	(d)	56.	(d)	57.	(b)	58.	(b)	59.	(c)	60.	(a)

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CALCULATION TECHNIQUES

TECHNIQUE - 01

♦ Cube Root of Non-perfect cube ♦

$$\sqrt[3]{N} = A \pm \frac{B}{3.A^2}$$

First find a number which is a perfect cube near to the number N, let it be A and the difference between N and A^3 is B.

Note: '+' sign is used if $N > A^3$ & '–' sign is used if $N < A^3$

Ex. $\sqrt[3]{60} = 4 - \frac{4}{3\times 4^2} = \frac{188}{48} = 3.92$

Ex. $\sqrt[3]{72} = 4 + \frac{8}{3\times 4^2} = \frac{200}{48} = 4.17$

♦ Do it Yourself ♦

(i) $\sqrt[3]{10}$ = ...

(ii) $\sqrt[3]{66}$ = ...

(iii) $\sqrt[3]{118}$ = ...

(iv) $\sqrt[3]{226}$ = ...

(v) $\sqrt[3]{525}$ = ...

(vi) $\sqrt[3]{1365}$ = ...

(vii) $\sqrt[3]{1500}$ = ...

TECHNIQUE - 02

♦ Square of two digit number ♦

+ Carry + Carry

$(ab)^2 = a^2/2ab/b^2$

1 digit, 1 digit

- **Square of 14**

Here, a = 1 & b = 4

$14^2 = 1^2/2.1.4/4^2 = 1/8/16 = 196$

- **Square of 23**

Here, a = 2 & b = 3

$23^2 = 2^2/2.2.3/3^2 = 4/12/9 = 529$

♦ Do it Yourself ♦

(i) 18^2 = ... (ii) 23^2 = ...

(iii) 27^2 = ... (iv) 34^2 = ...

(v) 38^2 = ... (vi) 43^2 = ...

(vii) 57^2 = ... (viii) 64^2 = ...

(ix) 66^2 = ... (x) 75^2 = ...

(xi) 78^2 = ... (xii) 84^2 = ...

SELF MENTORSHIP TABLE

	STEP-01	STEP-02	STEP-03	STEP-04	STEP-05	STEP-06
TOPIC NAME	VIDEOS	CLASSROOM SHEET	SSC MATHS	NOTES / FORMULA BOOK REVISION	CHAPTER TEST	PRO LEVEL PRACTICE SHEET
TRIGONOMETRY SHEET - 05						
MULTIPLE AND SUB-MULTIPLE TRIGONOMETRIC FUNCTION OF 2A						
TRIGONOMETRIC FUNCTION OF 3A						
IDENTITIES BASED ON SUM & DIFFERENCE OF ANGLE OF T-RATIO						
TRANSFORMATION OF SUM OR DIFFERENCE INTO PRODUCT						
TRANSFORMATION OF A PRODUCT INTO SUM OR DIFFERENCE						
SOME IMPORTANT RESULTS						